`Final Project` Calculation of Lake Pollutants

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May 30, 2025

Problem 1

The initial conditions are $C_1(0) = C_{10}$ and $C_2(0) = C_{20}$. Lake L_1 is affected by pollutant inflow and outflow. We assume that water with a flow rate of $Q\,\mathrm{m}^3/\mathrm{day}$ (with a pollutant concentration of $C_0\,\mathrm{g/m}^3$) flows into Lake L_1, and at the same time, water with the same flow rate $Q\,\mathrm{m}^3/\mathrm{day}$ flows out of Lake L_1 into Lake L_2. So that

$$\frac{\mathrm{d}(V_1 C_1(t))}{\mathrm{d}t} = QC_0(t) - QC_1(t)$$

Similarly, the equation plus external pollutant input, i.e., pollutants with a mass of W g discharged into Lake L_2 per day (unit: g/day) will be

$$\frac{\mathrm{d}(V_2 C_2(t))}{\mathrm{d}t} = QC_1(t) - QC_2(t) + W$$

Since $\beta_1 = \frac{Q}{V_1}$, $\beta_2 = \frac{Q}{V_2}$ and $\beta_3 = \frac{W}{V_2}$, we divide the equations by V_1 and V_2 respectively and then have

$$\begin{cases} \dfrac{\mathrm{d}C_1(t)}{\mathrm{d}t} = \beta_1\, C_0(t) - \beta_1\, C_1(t), \\ \dfrac{\mathrm{d}C_2(t)}{\mathrm{d}t} = \beta_2\, C_1(t) - \beta_2\, C_2(t) + \beta_3, \\ C_1(0) = C_{10}, \\ C_2(0) = C_{20}. \end{cases}$$

Problem 2

The solutions are:

$$
\begin{aligned}
C_1(t) &= C_0 - C_0 \cdot \exp(-\beta_1 t) + C_{10} \cdot \exp(-\beta_1 t) \\
C_2(t) &= \exp(-\beta_1 t)\left(\frac{\beta_2(C_0 - C_{10})}{\beta_1 - \beta_2} - \frac{C_0\beta_2\exp(\beta_1 t)}{\beta_1 - \beta_2}\right) \\
&\qquad - \exp(-\beta_2 t)\left(\frac{\beta_1\beta_3 - \beta_2\beta_3 - C_{10}\beta_2^2 + C_{20}\beta_2^2 + C_0\beta_1\beta_2 - C_{20}\beta_1\beta_2}{\beta_2(\beta_1 - \beta_2)}\right. \\
&\qquad \left. - \frac{\exp(\beta_2 t)(C_0\beta_1 - \beta_3 + \frac{\beta_1\beta_3}{\beta_2})}{\beta_1 - \beta_2}\right)
\end{aligned}
$$

```matlab
%--------------Problem 2--------------
clc; clear; close all;

% Define Variables
syms C1(t) C2(t) beta1 beta2 beta3 C0 C10 C20

% Define Equations
eq1 = diff(C1, t) == beta1 * (C0 - C1);
eq2 = diff(C2, t) == beta2 * (C1 - C2) + beta3;

% Define Initial Conditions
cond1 = C1(0) == C10;
cond2 = C2(0) == C20;

% Solve the ODE System
[C1_sol, C2_sol] = dsolve([eq1, eq2], [cond1, cond2]);

% Simplify the Solutions
C1_sol = simplify(C1_sol);
C2_sol = simplify(C2_sol);

% Display the Analytical Solutions
disp('C1(t) = ');
disp(C1_sol);
disp('C2(t) = ');
disp(C2_sol);

% Output:
% C1(t) =
% C0 - C0*exp(-beta1*t) + C10*exp(-beta1*t)
% C2(t) =
% exp(-beta1*t)*((beta2*(C0 - C10))/(beta1 - beta2) - ...
    (C0*beta2*exp(beta1*t))/(beta1 - beta2)) - ...
    exp(-beta2*t)*((beta1*beta3 - beta2*beta3 - C10*beta2^2 + ...
    C20*beta2^2 + C0*beta1*beta2 - C20*beta1*beta2)/(beta2*(beta1 ...
    - beta2)) - (exp(beta2*t)*(C0*beta1 - beta3 + ...
    (beta1*beta3)/beta2))/(beta1 - beta2))
```

Problem 3

We use the analytical solution to fit the model. Since there's only 1 parameter β_1 in the first equation, we adopt the two steps `lsqcurvefit`. The capacity of lake L_1 is estimated to be 200000.00 m^3 and the capacity of lake L_2 is estimated to be 100000.00 m^3.

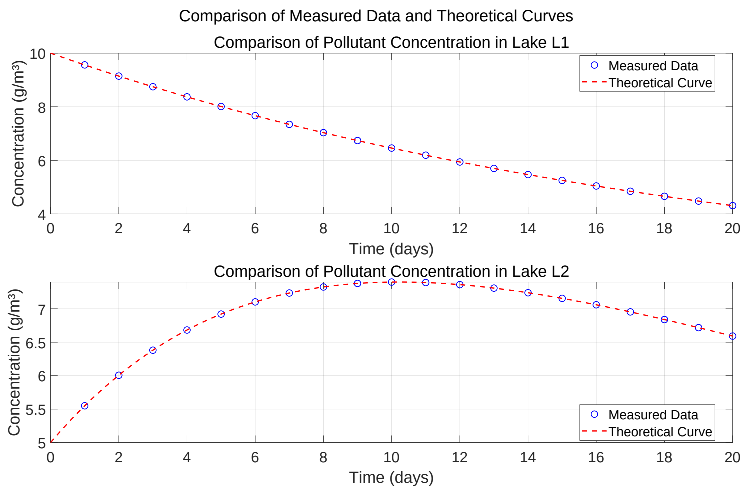

Figure 1: Problem 3

```
%-----------Problem 3-----------
clc; clear; close all;

% Data and Parameters
data = readmatrix('data.xls', 'NumHeaderLines', 1);
t_data  = data(:, 1);
C1_data = data(:, 2);
C2_data = data(:, 3);
Q = 1e4;      % m^3/day
C0 = 1;       % g/m^3
C10 = 10;     % g/m^3
C20 = 5;      % g/m^3
W = 1e4;      % g/day
K = C0 + W/Q; % = 2 g/m^3

%% Estimate V Sequentially
% Define V1 Model Function
model_C1 = @(beta1, t) C0 + (C10 - C0) * exp(-beta1 * t);
beta1_init = 0.1;  % Initial Guess Values
opts = optimoptions('lsqcurvefit', 'Display', 'off');
beta1_fit = lsqcurvefit(@(b,t) model_C1(b,t), beta1_init, t_data, ...
    C1_data, [], [], opts);
V1 = Q / beta1_fit;
```

```matlab

% Define V2 Model Function
model_C2 = @(beta2, t) ...
    (C20 - K - (beta2 * (C10 - C0)) / (beta2 - beta1_fit)) .* ...
        exp(-beta2 * t) + ...
    K + (beta2 * (C10 - C0) / (beta2 - beta1_fit)) .* ...
        exp(-beta1_fit * t);
beta2_init = 0.05;  % Initial Guess Values
beta2_fit = lsqcurvefit(model_C2, beta2_init, t_data, C2_data, ...
    [], [], opts);
V2 = Q / beta2_fit;

%% Display Estimated Results
fprintf('Estimated lake capacities:\n');
fprintf('V1 = %.2f m^3\n', V1);
fprintf('V2 = %.2f m^3\n', V2);

t_fit = linspace(0, max(t_data), 100);
C1_fit = model_C1(beta1_fit, t_fit);
C2_fit = model_C2(beta2_fit, t_fit);

%% Plot and Export
figure;
subplot(2, 1, 1);
plot(t_data, C1_data, 'bo', 'LineWidth', 1, 'MarkerSize', 8, ...
    'DisplayName', 'Measured Data');
hold on;
plot(t_fit, C1_fit, 'r--', 'LineWidth', 1.5, 'DisplayName', ...
    'Theoretical Curve');
title('Comparison of Pollutant Concentration in Lake L1', ...
    'FontSize', 12);
xlabel('Time (days)', 'FontSize', 10);
ylabel('Concentration (g/m³)', 'FontSize', 10);
legend('Location', 'best');
set(gca, 'FontSize', 18);
grid on;

subplot(2, 1, 2);
plot(t_data, C2_data, 'bo', 'LineWidth', 1, 'MarkerSize', 8, ...
    'DisplayName', 'Measured Data');
hold on;
plot(t_fit, C2_fit, 'r--', 'LineWidth', 1.5, 'DisplayName', ...
    'Theoretical Curve');
title('Comparison of Pollutant Concentration in Lake L2', ...
    'FontSize', 12);
xlabel('Time (days)', 'FontSize', 10);
ylabel('Concentration (g/m³)', 'FontSize', 10);
legend('Location', 'best');
set(gca, 'FontSize', 18);
grid on;

sgtitle('Comparison of Measured Data and Theoretical Curves', ...
    'FontSize', 20);
set(gcf, 'Position', [100, 100, 1600, 900]);
exportgraphics(gcf, 'Problem3.pdf', 'ContentType', 'vector');

% Output:
```

```
% Estimated lake capacities:
% V1 = 200000.00 m^3
% V2 = 100000.00 m^3
```

Problem 4

It takes 245 days for the concentration of pollutants in lake L_1 and lake L_2 to both be below 2 g/m^3.

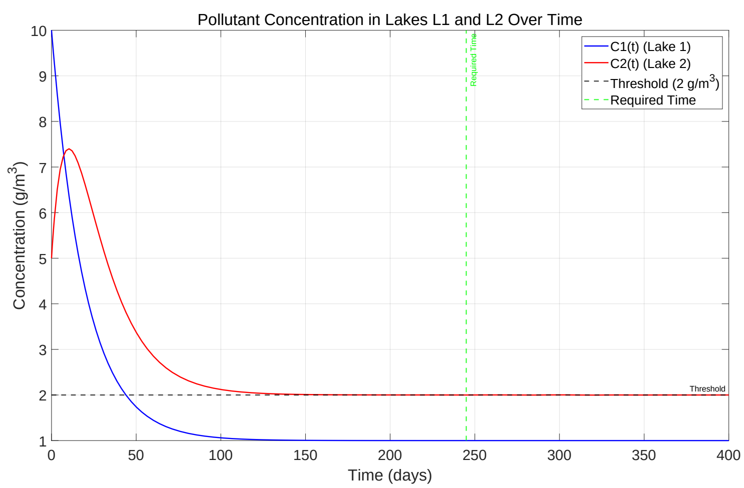

Figure 2: Problem 4

```
%-------------Problem 4-------------
clc; clear; close all;

%% Parameters and Coefficients
Q = 1e4;        % m^3/day
C0 = 1;         % g/m^3
C10 = 10;       % g/m^3
C20 = 5;        % g/m^3
W = 1e4;        % g/day
V1 = 200000.00;      % m^3
V2 = 100000.00;       % m^3

beta1 = Q / V1;
beta2 = Q / V2;
beta3 = W / V2;

%% Solve ODE
```

```
odefun = @(t, C) [beta1 * (C0 - C(1)); beta2 * (C(1) - C(2)) + ...
    beta3];

t_span = [0 400];
[t, C] = ode45(odefun, t_span, [C10; C20]);

% Find the Time Below 2 g/m^3
idx = find(C(:, 1) < 2 & C(:, 2) < 2, 1, 'first');
if ¬isempty(idx)
    time_required = t(idx);
    fprintf('It takes %.2f days for C1 and C2 to both be below 2 ...
        g/m^3\n', time_required);
else
    fprintf('The requirement is not met within 400 days\n');
end

%% Plotting and Save
figure;
plot(t, C(:, 1), 'b-', 'LineWidth', 1.5);
hold on;
plot(t, C(:, 2), 'r-', 'LineWidth', 1.5);
yline(2, 'k--', 'Threshold', 'LineWidth', 1.5);
if ¬isempty(idx)
    xline(time_required, 'g--', 'Required Time', 'LineWidth', 1.5);
end
xlabel('Time (days)');
ylabel('Concentration (g/m^3)');
title('Pollutant Concentration in Lakes L1 and L2 Over Time');
legend('C1(t) (Lake 1)', 'C2(t) (Lake 2)', 'Threshold (2 g/m^3)', ...
    'Required Time');
grid on;

set(gca, 'FontSize', 18);
set(gcf, 'Position', [100, 100, 1600, 900]);
exportgraphics(gcf, 'Problem4.pdf', 'ContentType', 'vector');

% Output:
% It takes 244.95 days for C1 and C2 to both be below 2 g/m^3
```

Problem 5

The target cannot be achieved within 40 days even for the maximum 15000.00 m^3/day. For Q = 15000.00 m^3/day, the target is achieved after 54 days.

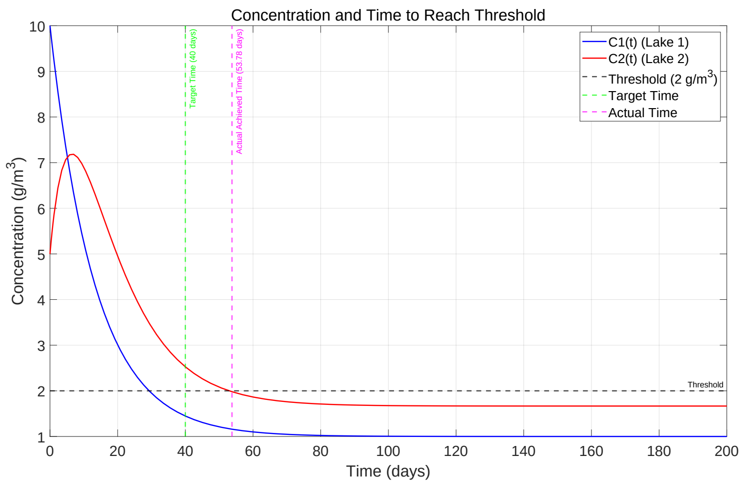

Figure 3: Problem 5

```
%---------------Problem 5---------------
clc; clear; close all;

%% Parameters
C0 = 1;         % g/m^3
C10 = 10;       % g/m^3
C20 = 5;        % g/m^3
W = 1e4;        % g/day
V1 = 200000.00;     % m^3
V2 = 100000.00;       % m^3
Q_max = 1.5e4;      % m^3/day
t_target = 40;      % days

%% Search for Q and Solve ODE
Q_values = linspace(0, Q_max, 100);
found = false;
Q_opt = Q_max;
t_required = NaN;

for Q = Q_values
    beta1 = Q / V1;
    beta2 = Q / V2;
    beta3 = W / V2;
    odefun = @(t, C) [beta1 * (C0 - C(1)); beta2 * (C(1) - C(2)) ...
        + beta3];
    [t, C] = ode45(odefun, [0 t_target], [C10; C20]);
    if C(end,1) < 2 && C(end,2) < 2
        fprintf('For Q = %.2f m^3/day, the target is achieved ...
            within 40 days\n', Q);
        Q_opt = Q;
        t_required = t(end);
        found = true;
        break
```

```
    end
end

if ¬found
    fprintf('The target cannot be achieved within 40 days for Q_max = %.2f m^3/day\n', Q_max);
    Q_opt = Q_max;
    beta1 = Q_opt / V1;
    beta2 = Q_opt / V2;
    beta3 = W / V2;
    odefun = @(t, C) [beta1 * (C0 - C(1)); beta2 * (C(1) - C(2)) + beta3];
    [t_long, C_long] = ode45(odefun, [0 200], [C10; C20]);

    idx = find(C_long(:,1) < 2 & C_long(:,2) < 2, 1);
    if isempty(idx)
        fprintf('Even after 200 days, concentrations do not fall below 2 g/m^3.\n');
        t_required = NaN;
    else
        t_required = t_long(idx);
        fprintf('For Q = %.2f m^3/day, the target is achieved after %.2f days.\n', Q_opt, t_required);
    end

    t = t_long;
    C = C_long;
end

%% Plotting and Save
figure;
plot(t, C(:, 1), 'b-', 'LineWidth', 1.5); hold on;
plot(t, C(:, 2), 'r-', 'LineWidth', 1.5);
yline(2, 'k--', 'Threshold', 'LineWidth', 1.5);
xline(t_target, 'g--', 'Target Time (40 days)', 'LineWidth', 1.5);
if ¬isnan(t_required)
    xline(t_required, 'm--', sprintf('Actual Achieved Time (%.2f days)', t_required), 'LineWidth', 1.5);
end
title('Concentration and Time to Reach Threshold');
xlabel('Time (days)');
ylabel('Concentration (g/m^3)');
legend('C1(t) (Lake 1)', 'C2(t) (Lake 2)', ...
    'Threshold (2 g/m^3)', 'Target Time', 'Actual Time');
grid on;

set(gca, 'FontSize', 18);
set(gcf, 'Position', [100, 100, 1600, 900]);
exportgraphics(gcf, 'Problem5.pdf', 'ContentType', 'vector');

% Output:
% The target cannot be achieved within 40 days for Q_max = 15000.00 m^3/day
% For Q = 15000.00 m^3/day, the target is achieved after 53.78 days.
```

Problem 6

First we use `moving average filtering` to process the data. Then we use `lsqcurvefit` the same way as before. Flow rate Q is estimated to be 14818.00 m^3/day. The capacity of lake L_1 is estimated to be 296626.84 m^3 and the capacity of lake L_2 is estimated to be 196002.95 m^3. The answer might be wrong, since the figure and the data are much the same as in the problems before, but the results turn out to be different. I doublecheck the value estimated in step 1 and find that $\beta_1 = 0.0500$ is the same, which indicates that the data smoothing process is correct. While in step 2, $\beta_2 = 0.0756$. If it is not a problem of data modification, it must be due to the introduction of the other parameter Q. It may be unreasonable to estimate two parameter just given 1 nonlinear equation.

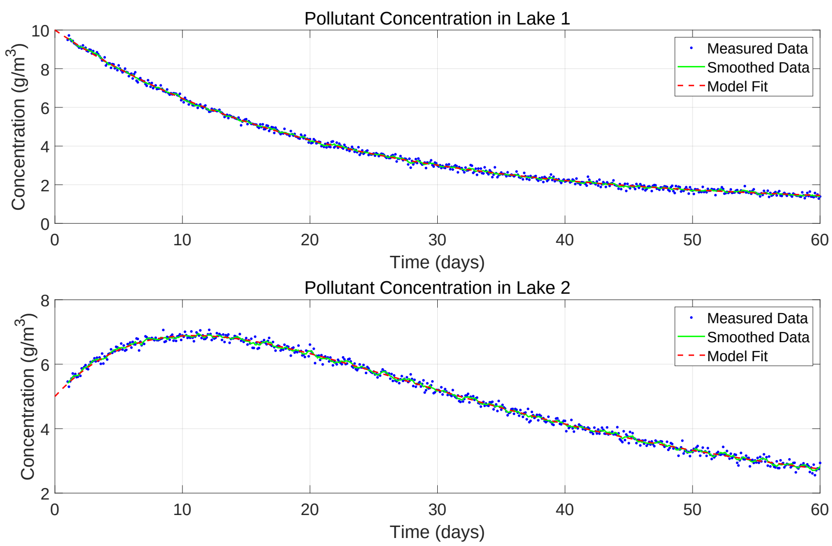

Figure 4: Problem 6

```
%-------------Problem 6-------------
clc; clear; close all;

%% Parameters
data = readmatrix('noisydata.xls', 'NumHeaderLines', 1);
t_data  = data(:, 1);
C1_data = data(:, 2);
C2_data = data(:, 3);
C0 = 1;        % g/m^3
C10 = 10;      % g/m^3
C20 = 5;       % g/m^3
W = 1e4;       % g/day

```

```matlab
%% Data Processing
window_size = 5;
C1_smooth = movmean(C1_data, window_size);
C2_smooth = movmean(C2_data, window_size);

%% Estimate V Sequentially
% Define V1 Model Function
model_C1 = @(beta1, t) C0 + (C10 - C0) * exp(-beta1 * t);
beta1_init = 0.05;  % Initial Guess Values
opts = optimoptions('lsqcurvefit', 'Display', 'off');
beta1_fit = lsqcurvefit(@(b,t) model_C1(b,t), beta1_init, t_data, ...
    C1_smooth, [], [], opts);

% Define V2 Model Function
model_C2 = @(para, t) ...
    (C20 - (C0 + W/para(2)) - (para(1) * (C10 - C0)) / (para(1) - ...
        beta1_fit)) .* exp(-para(1) * t) + ...
    C0 + W/para(2) + (para(1) * (C10 - C0) / (para(1) - ...
        beta1_fit)) .* exp(-beta1_fit * t);
para_init = [0.1, 1e4];  % Initial Guess Values
para_fit = lsqcurvefit(model_C2, para_init, t_data, C2_smooth, ...
    [], [], opts);
V1 = para_fit(2) / beta1_fit;
V2 = para_fit(2) / para_fit(1);

% Display Results
fprintf('Estimated Q = %.2f m^3/day\n', para_fit(2));
fprintf('Estimated V1 = %.2f m^3\n', V1);
fprintf('Estimated V2 = %.2f m^3\n', V2);

% Calculate Model Predictions
t_fit = linspace(0, max(t_data), 100);
C1_fit = model_C1(beta1_fit, t_fit);
C2_fit = model_C2(para_fit, t_fit);

%% Plotting and Saving
figure;
subplot(2,1,1);
plot(t_data, C1_data, 'b.', 'MarkerSize', 8);
hold on;
plot(t_data, C1_smooth, 'g-', 'LineWidth', 1.5);
plot(t_fit, C1_fit, 'r--', 'LineWidth', 1.5);
title('Pollutant Concentration in Lake 1');
xlabel('Time (days)');
ylabel('Concentration (g/m^3)');
legend('Measured Data', 'Smoothed Data', 'Model Fit');
set(gca, 'FontSize', 18);
grid on;

subplot(2,1,2);
plot(t_data, C2_data, 'b.', 'MarkerSize', 8);
hold on;
plot(t_data, C2_smooth, 'g-', 'LineWidth', 1.5);
plot(t_fit, C2_fit, 'r--', 'LineWidth', 1.5);
title('Pollutant Concentration in Lake 2');
xlabel('Time (days)');
ylabel('Concentration (g/m^3)');
```

```
legend('Measured Data', 'Smoothed Data', 'Model Fit');
set(gca, 'FontSize', 18);
grid on;

set(gcf, 'Position', [100, 100, 1600, 900]);
exportgraphics(gcf, 'Problem6.pdf', 'ContentType', 'vector');

% Output:
% Estimated Q = 14818.00 m^3/day
% Estimated V1 = 296626.84 m^3
% Estimated V2 = 196002.95 m^3
```